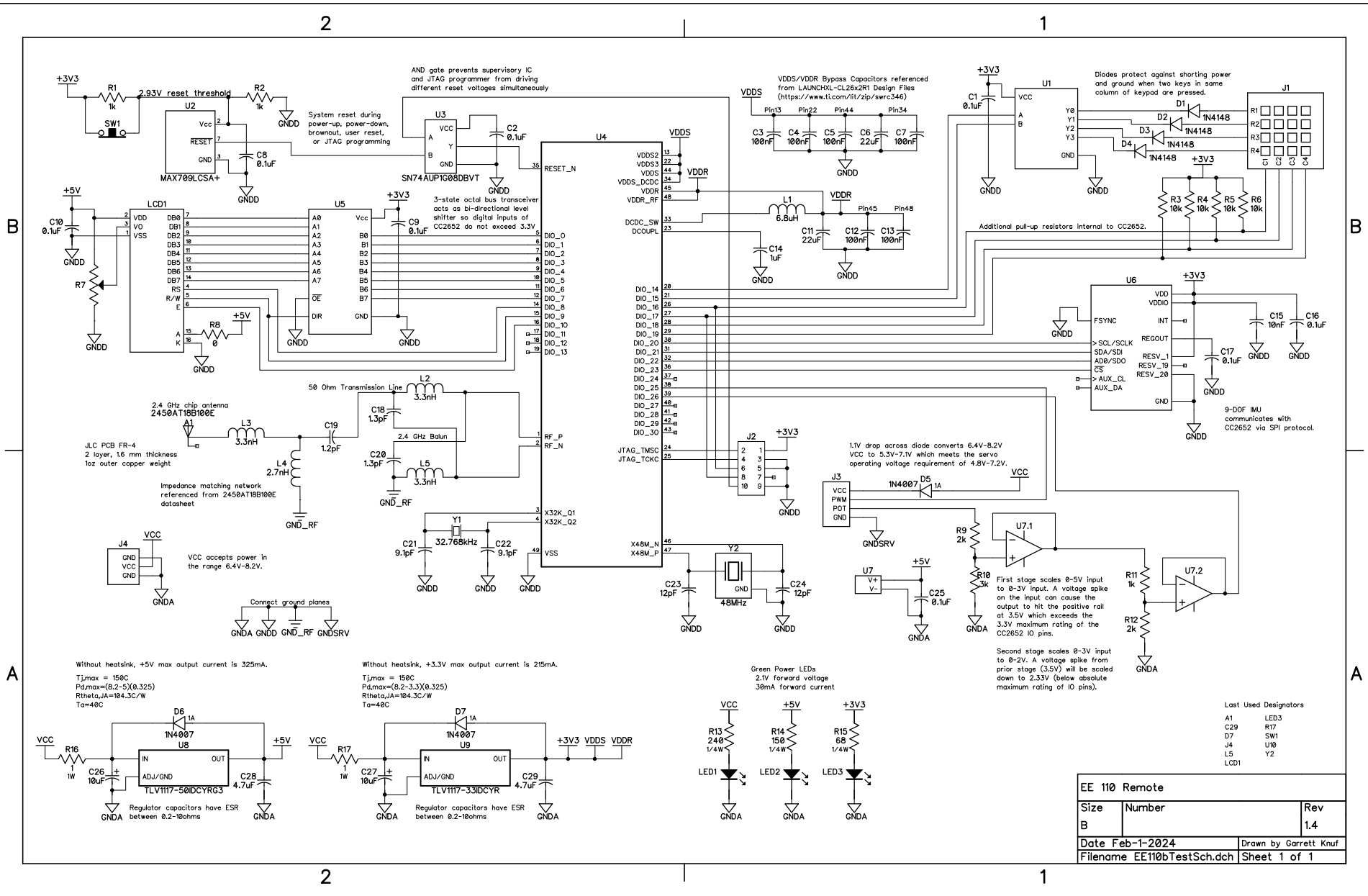

AND gate prevents supervisory IC and JTAG programmer from driving different reset voltages simultaneously
VDDS/VDDR Bypass Capacitors referenced from LAUNCHXL-CL26x2R1 Design Files (https://www.ti.com/lit/zip/swrc346)
Diodes protect against shorting power and ground when two keys in same column of keypad are pressed.
2.93V reset threshold
System reset during power-up, power-down, brownout, user reset, or JTAG programming
MAX709LCSA+
SN74AUP1G08DBVT
3-state octal bus transceiver acts as bi-directional level shifter so digital inputs of CC2652 do not exceed 3.3V
Additional pull-up resistors internal to CC2652.
9-DOF IMU communicates with CC2652 via SPI protocol.
50 Ohm Transmission Line
2.4 GHz chip antenna
2450AT18B100E
JLC PCB FR-4
2 layer, 1.6 mm thickness
1oz outer copper weight
2.4 GHz Balun
Impedance matching network referenced from 2450AT18B100E datasheet
32.768kHz
48MHz
VCC accepts power in the range 6.4V-8.2V.
Connect ground planes
1.1V drop across diode converts 6.4V-8.2V VCC to 5.3V-7.1V which meets the servo operating voltage requirement of 4.8V-7.2V.
First stage scales 0-5V input to 0-3V input. A voltage spike on the input can cause the output to hit the positive rail at 3.5V which exceeds the 3.3V maximum rating of the CC2652 IO pins.
Second stage scales 0-3V input to 0-2V. A voltage spike from prior stage (3.5V) will be scaled down to 2.33V (below absolute maximum rating of IO pins).
Without heatsink, +5V max output current is 325mA.
Tj,max = 150C
Pd,max=(8.2-5)(0.325)
Rtheta,JA=104.3C/W
Ta=40C
TLV1117-50IDCYRG3
Regulator capacitors have ESR between 0.2-10ohms
Without heatsink, +3.3V max output current is 215mA.
Tj,max = 150C
Pd,max=(8.2-3.3)(0.325)
Rtheta,JA=104.3C/W
Ta=40C
TLV1117-33IDCYR
Regulator capacitors have ESR between 0.2-10ohms
Green Power LEDs
2.1V forward voltage
30mA forward current
Last Used Designators
A1 LED3
C29 R17
D7 SW1
J4 U10
L5 Y2
LCD1
EE 110 Remote
Size B
Number
Rev 1.4
Date Feb-1-2024
Drawn by Garrett Knuf
Filename EE110bTestSch.dch
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